CASE STUDY

Olist E-commerce Analysis

Delivery — not price, category, or payment — is the single driver of customer satisfaction.

Poi
github.com/workintechpoyrazaka-sketch/olist-ecommerce-analysis
April 2026

THESIS

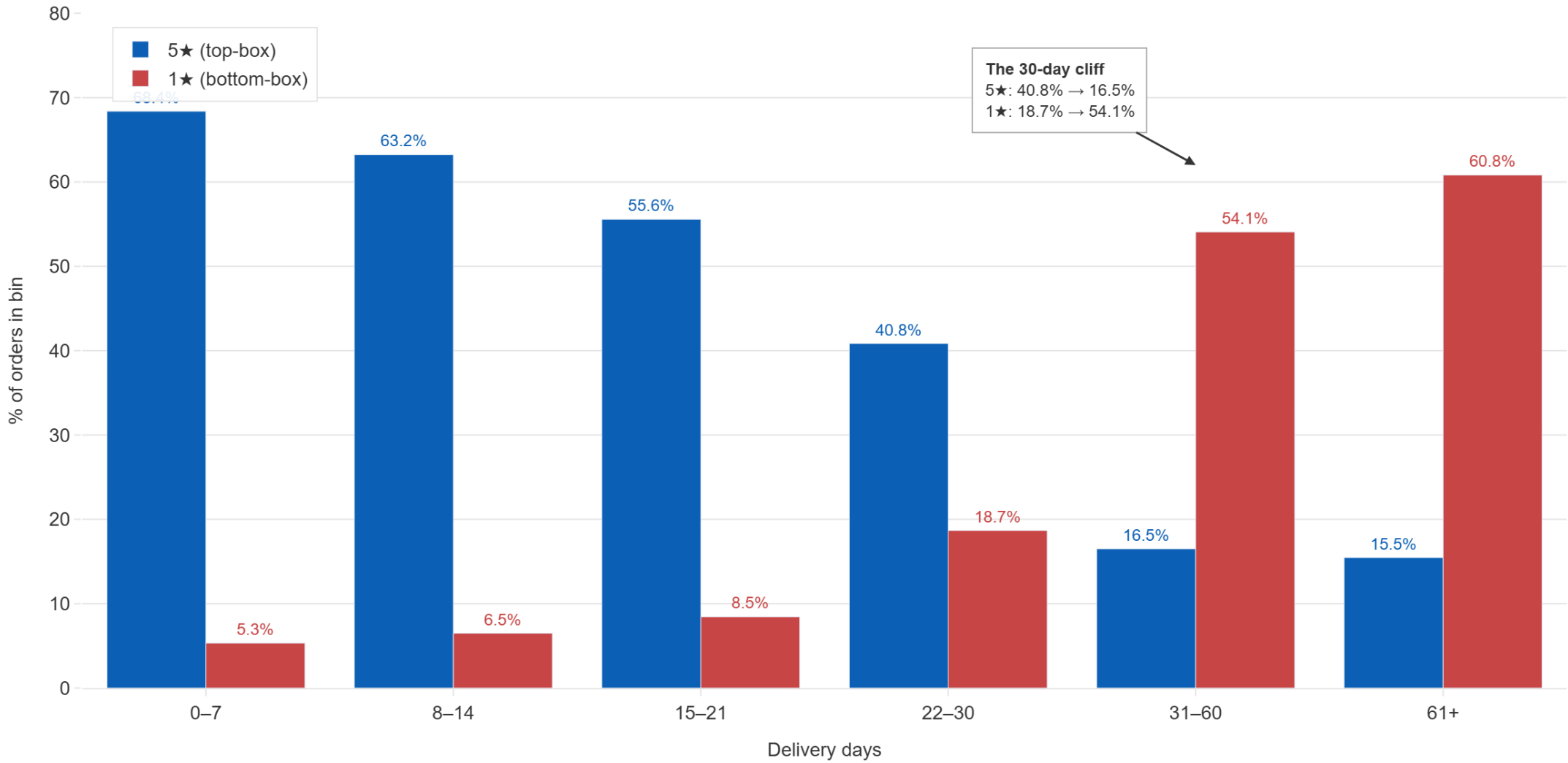
Olist satisfaction is delivery-driven.

Price, category, and payment type each produce trivial or null effects on review score — despite strong between-category price heterogeneity.

The one operational lever is on-time percentage — not average delivery speed.

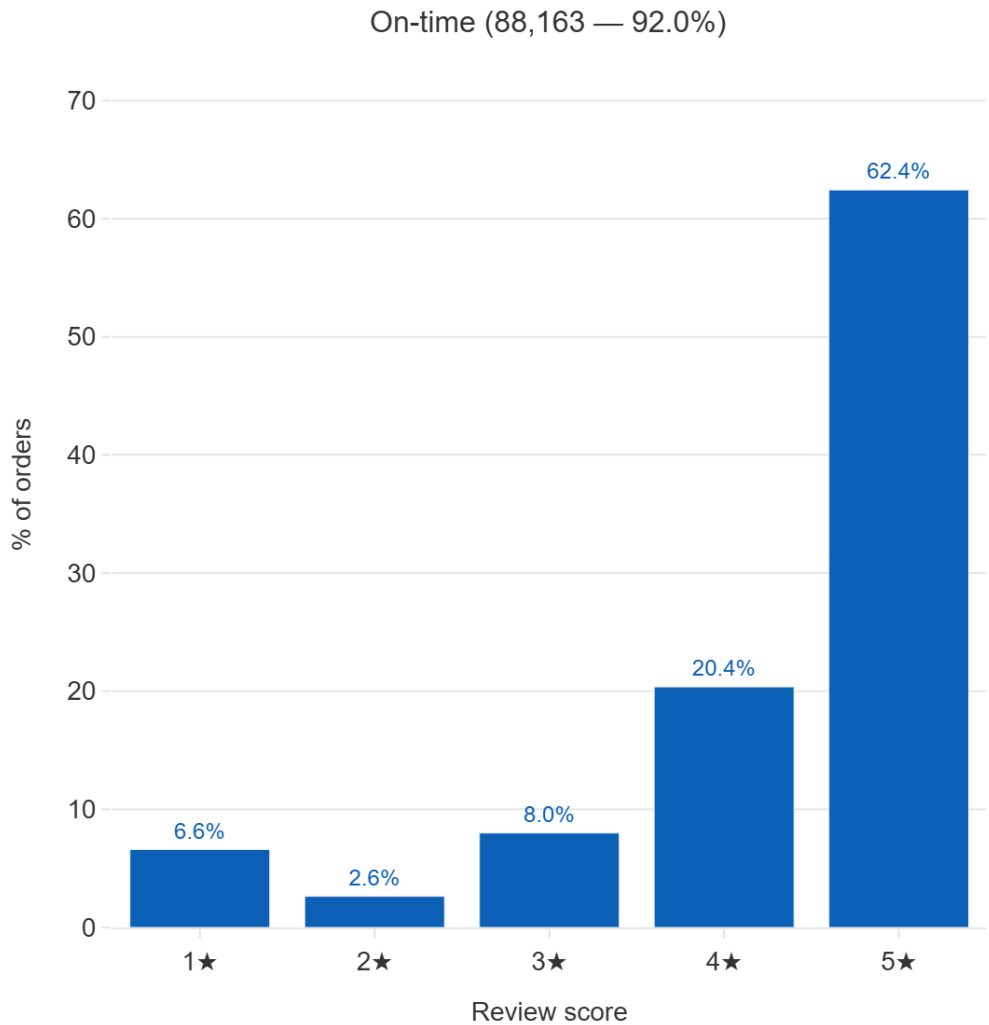
Satisfaction doesn't decay — it collapses at day 30

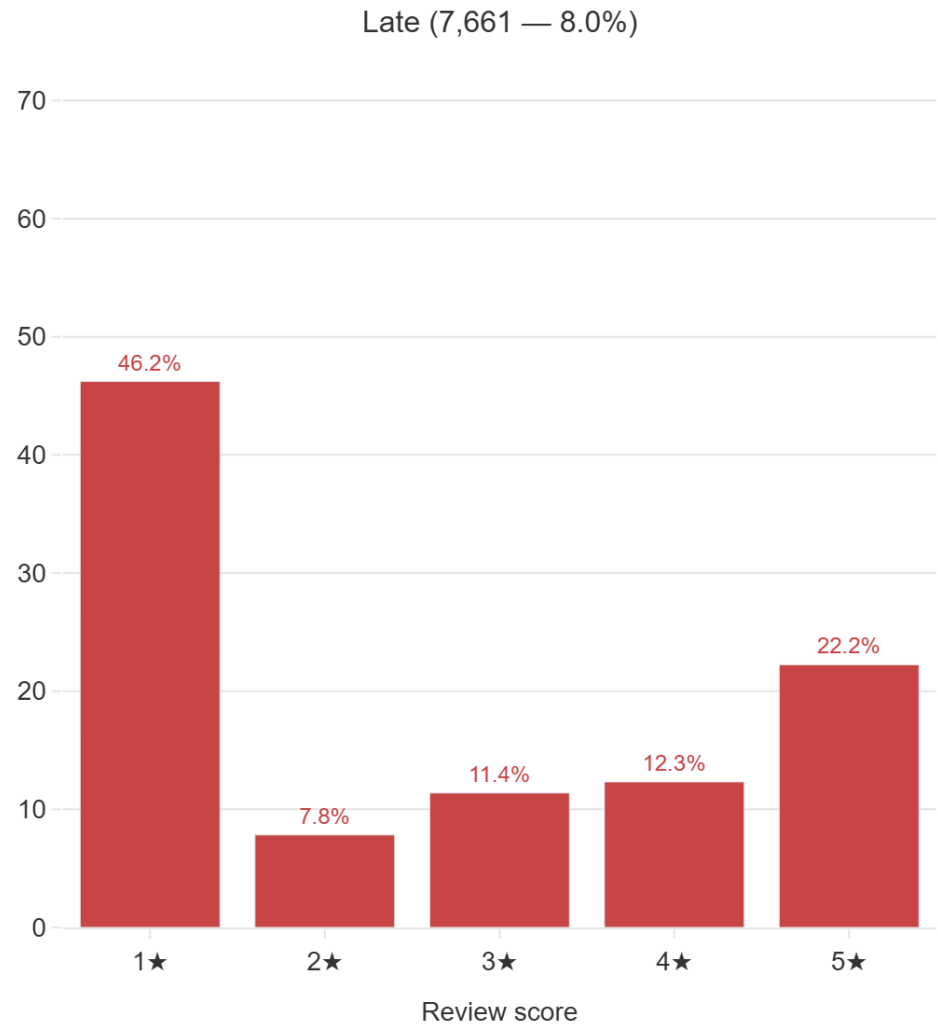
n = 95,824 delivered orders, Jan 2017 – Aug 2018



On-time delivery is the single operational lever for satisfaction

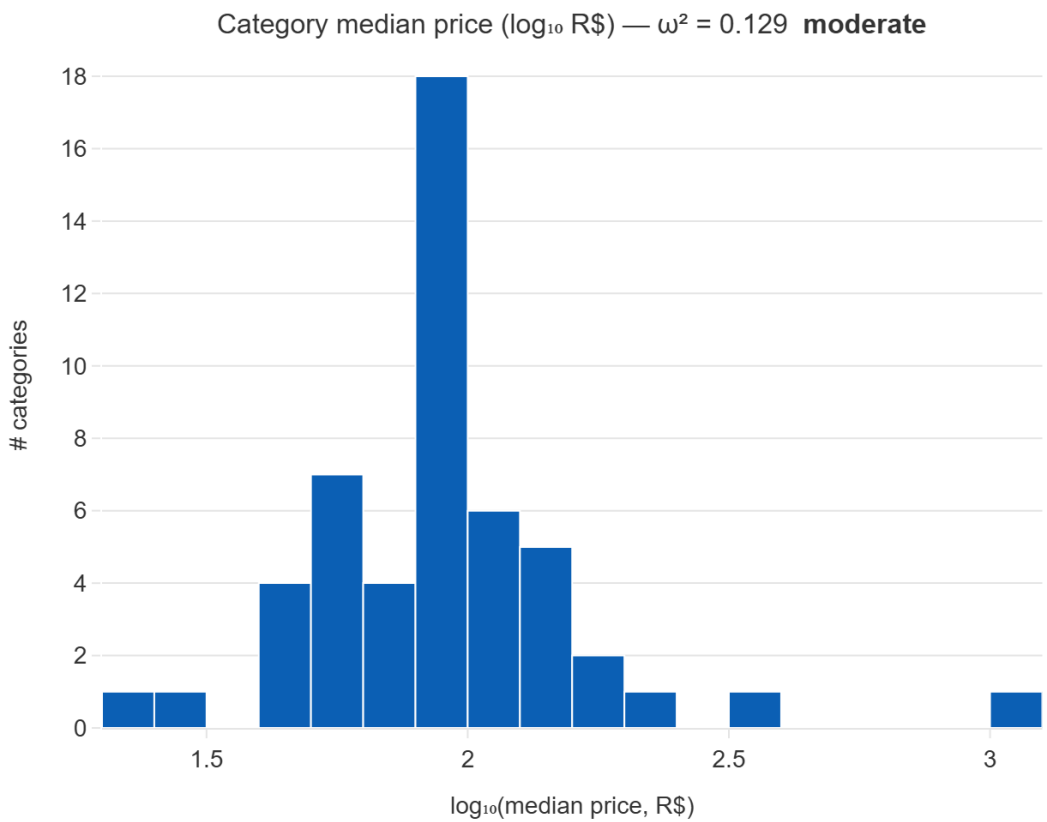
OR = 12.2× for 1★ reviews | +39.6pp bottom-box gap | rank-biserial |r| = 0.55

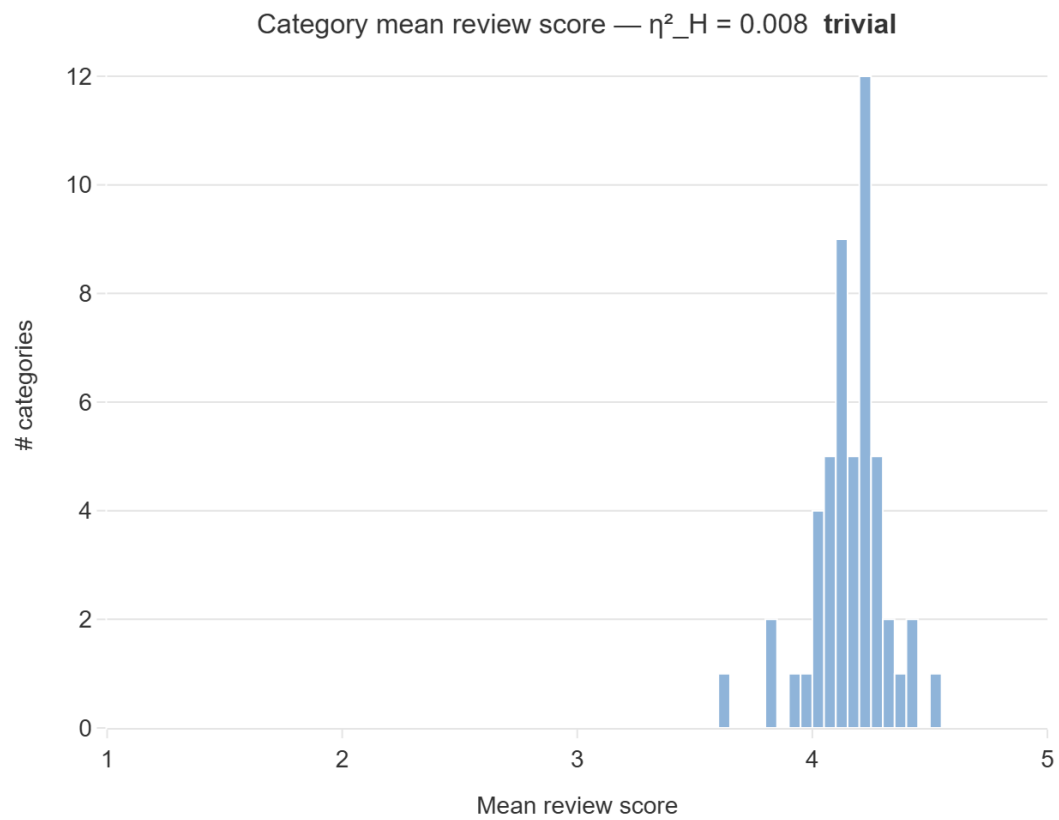




Categories disagree on what things cost, not on how satisfied customers are

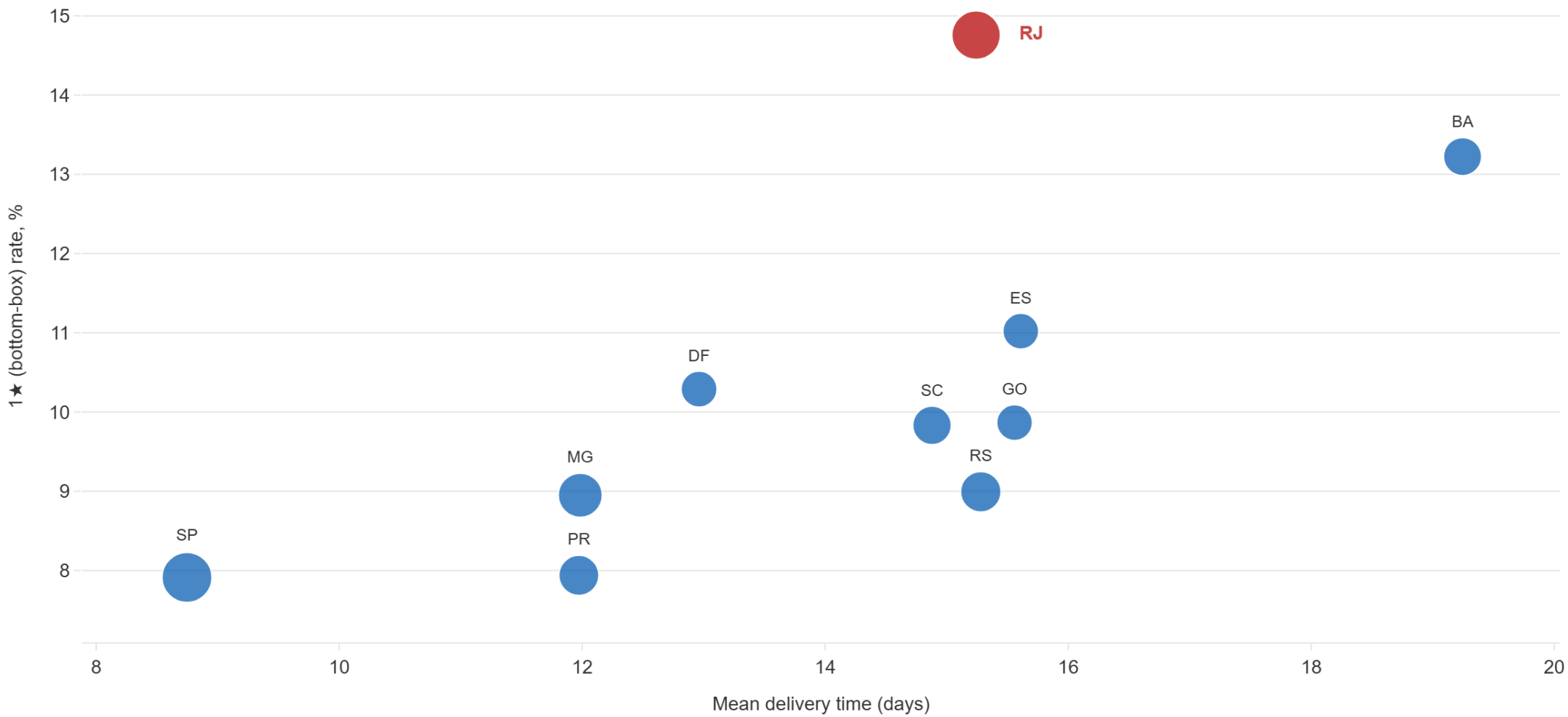
51 categories (n ≥ 100) | price ranges 55× across catalog | review means cluster around 4





State satisfaction is delivery-mediated, not regional

Spearman ρ = 0.76 across top 10 states | RJ: ~5pp higher 1★ rate than delivery time predicts



METHODOLOGY

How the analysis was done

~96k

delivered orders

Jan 2017 – Aug 2018, 20 full months

99.3%

review coverage

Flat across status, month, delivery speed

9

statistical tests

Spearman, Kruskal-Wallis, Mann-Whitney, Welch's, χ^2, z

Key decisions

Review score = ordinal.
Top-box (5★) and bottom-box (1★) rates instead of means.

Delivered-only filter.
Satisfaction tests restricted to completed deliveries.

Effect size > p-value.
At n≈96k, significance is near-universal.

Bootstrap CIs on ρ.
2000 resamples, paired, percentile, seed=42.

Min-n filter, 100.
20 tiny categories dropped — 0.8% of sample.

Cohen effect thresholds.
Trivial <0.10, weak <0.30, moderate <0.50, strong ≥0.50.

LIMITATIONS

What this analysis can and can't claim

01

Observational, not causal.

The analysis identifies correlations and consistent mediation patterns. It does not rule out unmeasured confounders.

02

Reviews dated before delivery (~8.5%) were retained, not filtered.

These represent customers submitting dispute reviews while still awaiting late shipments — the most dissatisfied segment. Removing them would bias top-box rates upward and systematically understate the delivery-satisfaction relationship.

03

Analytical window: Jan 2017 – Aug 2018.

Excludes late 2016 (low coverage) and Sep 2018 onward (partial month). Black Friday 2017 retained; coverage verified flat.

04

Minimum-n = 100 for category-level tests.

Cleaner rank-statistics and Dunn's matrix at the cost of 0.8% sample coverage. Trade-off documented in notebook 03.

WHAT I'D DO NEXT

Three investigations this data opens up

office_furniture

Retention dive

The only category combining top-5 median price with bottom-5 satisfaction. Plausible mechanisms — bulk-shipping damage, assembly complexity, expectation-price mismatch — cannot be confirmed from order data alone. Would require review-text NLP or a targeted customer survey.

Rio de Janeiro

Last-mile investigation

RJ sits ~5pp above the delivery-time trend line across top-10 states. Same mean delivery as SC/GO/RS, but measurably worse satisfaction. Points to a carrier-handoff or final-mile problem, not a logistics-distance problem.

Voucher orders

Composition audit

Voucher completion runs 4.7pp below other payment methods at 1.6% of orders. Most likely explanation is customer-intent composition (promo-driven buyers), not payment-infrastructure failure — but the intent data would need to be joined.

THANK YOU

Questions welcome.

If a finding stuck — or didn't — I want to know which one.

GitHub	github.com/workintechpoyrazaka-sketch/olist-ecommerce-analysis
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APPENDIX A

Full statistical results

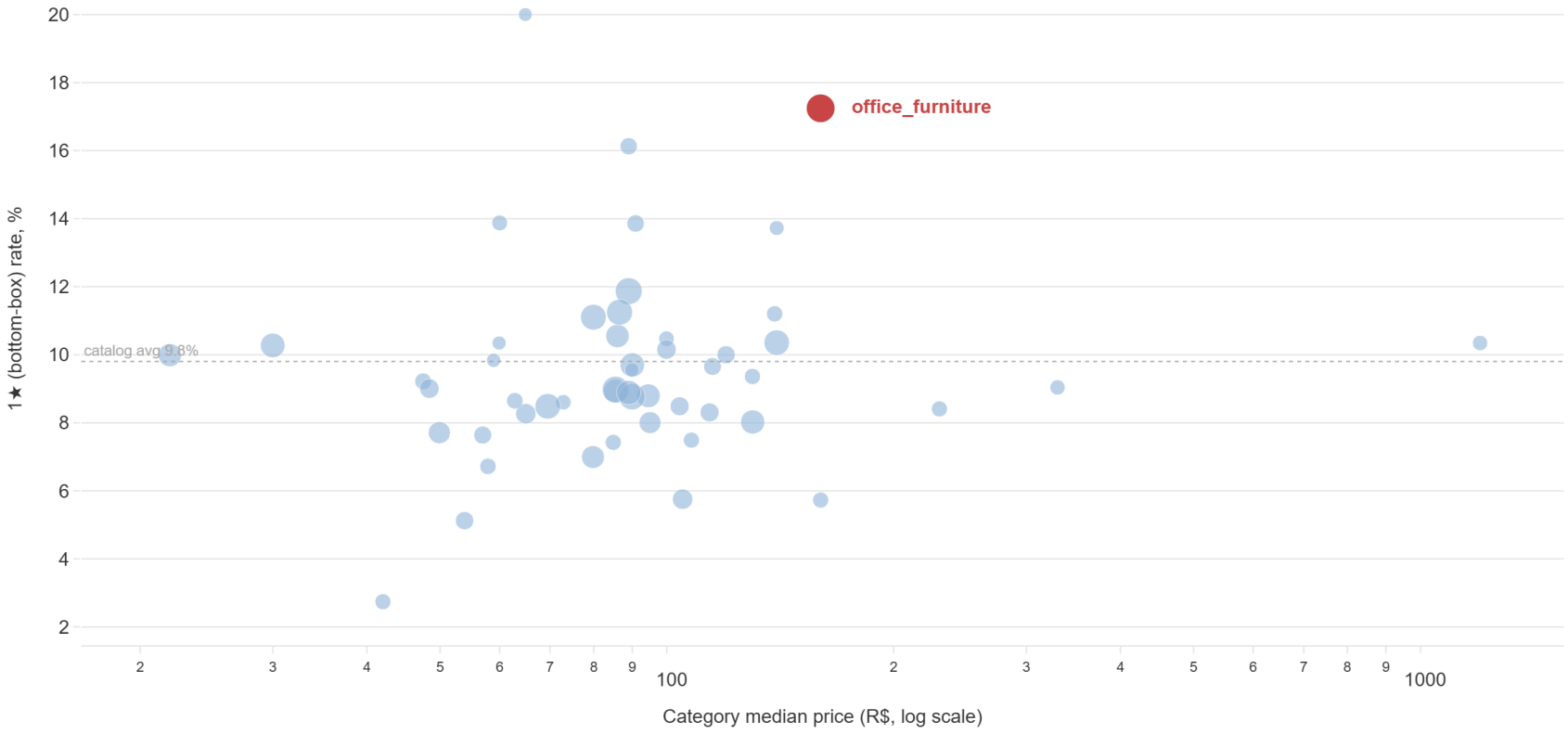
#	Relationship	Test	Result & effect size	Verdict
Q1	price → delivery	Spearman ρ	ρ = +0.102, CI [+0.095, +0.108]	weak
§4-A	delivery → review	Spearman ρ	ρ = −0.235 — 30-day cliff	**headline**
§4-B	on-time → review	Mann-Whitney + 2-prop z	|r| = 0.55; OR = 12.2× [11.6, 12.8]	**headline**
Q2	price → review	Spearman ρ	ρ = −0.029 (mediation consistent)	trivial
Q3	category → review	Kruskal-Wallis + Dunn's	η^2_H = 0.008; 186/1275 pairs sig	trivial
Q4a	payment → completion	Chi-square + Cramer's V	V = 0.035; voucher −4.7pp	trivial
Q4b	payment → review	Kruskal-Wallis	$\eta^2_H \approx 0.000$; p = 0.060	trivial
§1	category → $\log_{10}$(price)	Welch's ANOVA + Games-Howell	ω^2 = 0.129; 899/1275 pairs sig	**moderate**

APPENDIX B

office_furniture — the one category combining high price and high 1★ rate

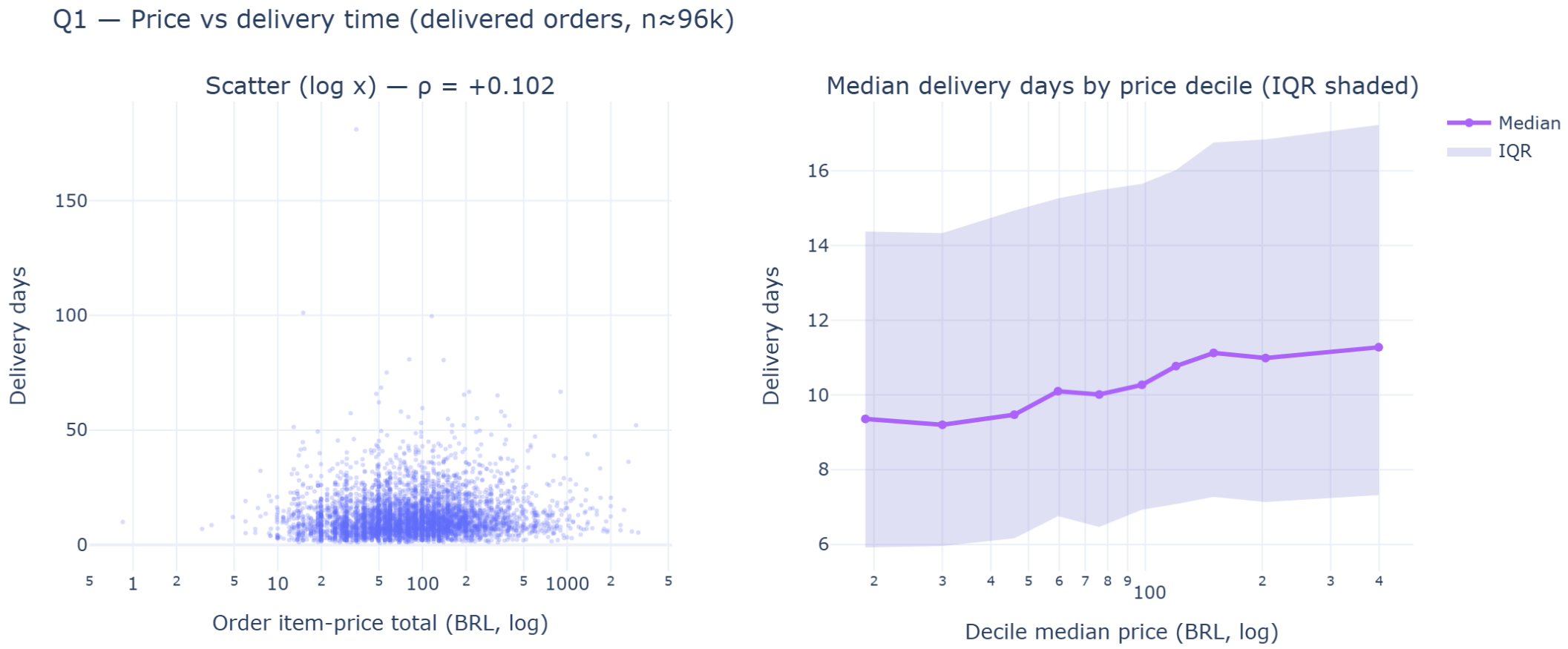
office_furniture: expensive and unsatisfying — the only category combining both

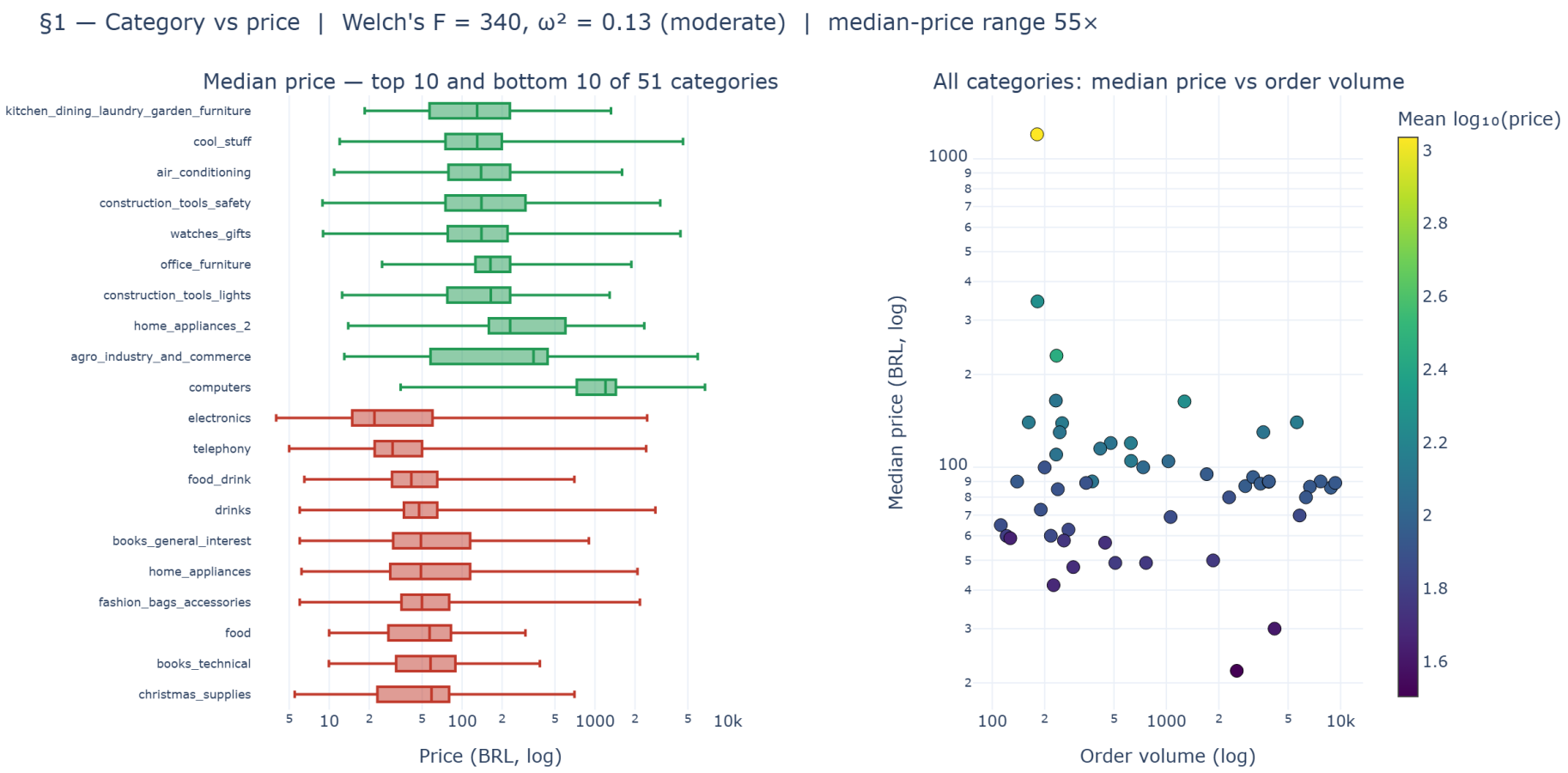
Top-5 median price AND bottom-5 satisfaction | n = 1,270 orders | 17.2% 1★ rate vs 9.8% catalog average



APPENDIX C

Q1 — Price vs delivery time (weak positive, ρ = +0.102)



APPENDIX D

§1 — Category explains price (ω² = 0.129, moderate)